

Test Case TC-001

Title: Basic city search returns listing cards

Priority: P0

Type: Functional

Preconditions: User is on the home page

Step #	Action	Expected Result
1	Enter “Dallas, TX” in search input	System accepts input
2	Submit search	System navigates to results page
3	Observe result panel	At least one listing card is displayed
4	Inspect card content	Listing contains mandatory fields (price, bedrooms, location)
5	Inspect page header	Header reflects “Dallas, TX” context
6	Observe browser console	No JavaScript errors displayed

Test Case TC-002

Title: Invalid location search shows clear guidance

Priority: P0

Type: Negative

Preconditions: User is on the home page

Step #	Action	Expected Result
1	Enter invalid string “asdf12345” in search input	System accepts input
2	Submit search	Results page loads without crashing
3	Inspect results panel	“No results” or guidance message displayed

4	Observe listings section	No stale or unrelated listings displayed
5	Observe UI consistency	User is guided to adjust search

Test Case TC-003

Title: Price filter minimum value is enforced

Priority: P0

Type: Boundary

Preconditions: User has performed city search with large inventory

Step #	Action	Expected Result
1	Open price filter	Filter UI opens
2	Set minimum price to 400,000	Value accepted in input field
3	Apply filter	Results refresh without errors
4	Inspect listings	No listing shows price below 400,000
5	Inspect filter chips	“Min \$400K” chip is displayed
6	Navigate via pagination	Filter remains applied across pages

Test Case TC-004

Title: Price filter maximum value is enforced

Priority: P0

Type: Boundary

Preconditions: City search with large inventory completed

Step #	Action	Expected Result
1	Open price filter	Filter UI opens
2	Set maximum price to 400,000	Value accepted
3	Apply filter	Results refresh
4	Inspect listings	No listing shows price above 400,000

5	Inspect filter chips	"Max \$400K" chip displayed
6	Refresh page	Filter persists post-refresh

Test Case TC-005

Title: Invalid price range (min > max) is prevented

Priority: P1

Type: Negative

Preconditions: User on city results page

Step #	Action	Expected Result
1	Open price filter	Filter component opens
2	Set min price to 600,000	System accepts input
3	Set max price to 400,000	System accepts input
4	Attempt to apply filter	Filter blocked by validation
5	Observe validation messaging	User informed min cannot exceed max
6	Attempt to bypass warning	No inconsistent results are displayed

Test Case TC-006

Title: Bedrooms filter enforces minimum bedroom count

Priority: P0

Type: Functional

Preconditions: City search with diverse bedroom inventory

Step #	Action	Expected Result
1	Open bedrooms filter	Filter opens
2	Select "3+ bedrooms"	Selection confirmed in UI

3	Apply filter	Results refresh
4	Inspect listings	No listing below 3 bedrooms displayed
5	Inspect chips	“3+ beds” chip visible
6	Refresh page	Filter persists across reload

Test Case TC-007

Title: Combined filters narrow results (price + bedrooms)

Priority: P0

Type: Functional

Preconditions: City search with substantial inventory

Step #	Action	Expected Result
1	Apply price filter 300,000–500,000	Filter accepted
2	Apply bedrooms filter “3+”	Filter accepted
3	Inspect listings	Listings satisfy both constraints
4	Compare result count vs baseline	Count reduced vs unfiltered search
5	Clear one filter	Only corresponding constraint removed
6	Clear all filters	Result set returns to baseline

Test Case TC-008

Title: Pagination preserves search context and filters

Priority: P0

Type: Functional

Preconditions: Filters applied and multi-page results available

Step #	Action	Expected Result
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1	Perform city search	Listings displayed
2	Apply price + bedroom filters	Filter UI confirms both active
3	Scroll to bottom	Pagination control visible
4	Navigate to next page	Filters remain active, listings still filtered
5	Navigate back	Context persists without reset
6	Observe duplicates	No unintended duplication across pages

Test Case TC-009

Title: No-results scenario handled gracefully

Priority: P1

Type: UX/Functional

Preconditions: City results page is active

Step #	Action	Expected Result
1	Apply extremely restrictive price filter	System accepts filter
2	Apply restrictive bedrooms filter	System accepts filter
3	Observe results	"No results" message displayed
4	Inspect listings area	No stale listings shown
5	Inspect UI guidance	Suggests broadening search

Test Case TC-010

Title: Clearing filters restores baseline results

Priority: P1

Type: Functional

Preconditions: User has active filters applied

Step #	Action	Expected Result
1	Verify price + beds filters active	Filters visible as chips
2	Click “Clear All Filters”	Filters removed from UI
3	Observe listings	Baseline unfiltered results displayed
4	Refresh page	State returns without filters
5	Inspect UI	No residual filter artifacts visible

Charter ID: EX-001

Title: Robustness of Listing Search with Combined Filters and Pagination.

Timebox: 60 minutes

Environment: Production site, Chrome desktop, normal network conditions.

Feature Under Test: Listing Search

Tester: Andrey Aguilar Chaves

1. Goal

Explore how the Listing Search feature behaves when users combine location search with price and bedroom filters, then navigate through multiple pages of results. Identify inconsistencies, defects, data mismatches, or UX gaps that may not surface in structured test cases.

2. Scope of Exploration

Focus Areas:

- Search for input validation (city, neighborhood, ZIP).
- Autocomplete suggestions (if presented).
- Price filter (min/max/open-ended).
- Bedrooms filter (min or “X+”).

- Interaction between search and filters.
- State consistency across pagination (“Next page” / “Load more”).
- No results in edge cases and messaging clarity.
- Browser back navigation behavior.
- Visible mandatory listing card fields.
- Unexpected data (missing fields, null, extreme values).

Out of Scope for this Charter:

- Account features (favorites, save search).
- Mobile/device matrix.
- Performance load testing.
- Accessibility of audits beyond minor observations.

3. Data and Locations for Exploration

Suggested high inventory markets for meaningful pagination and filtering:

- Dallas, TX.
- Houston, TX.
- Phoenix, AZ.
- Atlanta, GA.

Filters to explore:

- Minimum price (e.g., 300k+).
- Maximum price (e.g., 500k).
- Bedrooms (e.g., 2+, 3+, 4+ bedrooms).

Boundary experiments:

- Min > max (invalid price range).
- Very high, minimum price.
- Very high bedrooms count.
- Zero results in combinations.

4. Heuristics and Oracles Used

SFDPOT Heuristic:

- Structure: Page layout, search bar, filters, cards, pagination.
- Function: Search, filter, paginate, update state.
- Data: Listing content fields, price, bedroom counts.
- Platform: Chrome browser, desktop environment.
- Operations: Rapid filter changes, deep pagination, back navigation.
- Time: Data changing between searches due to real inventory.

Consistency Oracles:

- UI behavior should match typical e-commerce filtering patterns.
- Filters should narrow results, not broaden them.
- Pagination should preserve the state.
- No results should be clearly communicated.
- No technical errors should surface to users.

5. Risks and Observations to Target

Potential Issues:

- Filters do not apply to future pages.
- Filters lost on browser back navigation.
- Wrong results due to min/max price inversion.
- Listings missing mandatory fields (e.g., price, bedrooms).
- Pagination duplicating listings.
- No results flow, not intuitive.
- Slow rendering or blocking UI states.
- Unexpected JavaScript or network errors.

6. Artifacts and Deliverables.

During this charter, the tester will produce:

- At least one formal bug report.
- Screenshot evidence of observed issues.
- Notes on potential data inconsistencies.

- Improvement suggestions for UX, quality, or testability.

7. Exit Criteria

Charter concludes when:

- Timebox (60 minutes) expires, OR
- All exploratory questions have been sufficiently exercised.

Defect Report (Sample)

Bug ID: BUG-SEARCH-014

Title: Price and bedroom filters are missing when navigating to page 2

Environment: Production – <https://www.newhomesource.com>, Chrome on Windows 11

Severity: High

Priority: P0

Status: New

Steps to Reproduce:

1. Navigate to the home page.
2. Search for “Dallas, TX”.
3. Apply price range 300k–500k.
4. Apply bedrooms filter “3+”.
5. Scroll to bottom and click the page 2 or the right arrow.
6. Observe the filter chips and listing results on page 2.

Expected Result:

Filters remain applied across pagination. Page 2 listings must also match the same price and bedroom constraints.

Actual Result:

Filters visually disappear or reset on page 2. Listings displayed include homes outside the expected price range and bedroom count. Some homes below 300k are shown.

Evidence:

The screenshot displays the NewHomeSource website interface for searching homes in Seattle, WA. The search filters at the top are set to "Seattle, WA", "Price: \$200k-\$1.3M", "3+ Br/Bath", and "Move-in date". The results are categorized under "New Homes in Seattle, WA" and "210 Communities". The listings include:

- Mill Bridge**: From \$649,950. 3-5 Beds, 2-3 Baths, 1,870 - 3,165 SqFt. Community in Lake Stevens, WA 98258. By 40 Homes. \$4.5K.
- Senderos**: From \$999,999. 3-4 Beds, 3 Baths, 2,000 - 2,002 SqFt. Community in Kirkland, WA 98034. By Taper Homes.
- Altamira**: From \$1,054,900. 3-5 Beds, 2 Baths, 2,437 - 2,701 SqFt. Community in Seattle, WA 98146. By Corner Homes.
- Salmon Creek Townhomes**: From \$689,950. 3-4 Beds, 2 Baths, 1,604 - 1,903 SqFt. Community in Seattle, WA 98146. By Blue Bear Homes.
- Gardens at Skyway**: From \$699,900. 2-3 Beds, 2-3 Baths, 1,384 - 1,987 SqFt. Community in Seattle, WA 98119. By Terra Homes.
- Skyway Village**: From \$979,900. 3-4 Beds, 3 Baths, 2,088 - 3,240 SqFt. Community in Seattle, WA 98119. By Terra Homes.

The page also shows "Communities near Seattle, WA" with additional listings like "Ten Trails Mountain View - To...", "Surprise Lake", "Eaglemont", "Garibaldi", "Regency at Ten Trails - Horizon...", "Regency at Ten Trails - Solstice...", "Regency at Ten Trails - Nova C...", "Regency at Ten Trails - Eclipse...", "Olympic Terrace", and "Falcon Ridge".

Suggested Fix Hint:

Ensure pagination of requests to include active filter parameters in the API call or query string. Preserve filter state in UI state management.

DATA VALIDATION (ASSUMED SCHEMA)

Assumed Database Tables:

Table: Listings

Columns:

ListingID (PK)

City

State

Price

Bedroom

Bathrooms

IsActive

LastUpdated

BuilderID

SQL Query 1

Goal: Confirm no listings returned for a given search to violate the UI-applied price filter.

```
SELECT ListingID, City, Price, Bedrooms
```

```
FROM Listings
```

```
WHERE City = 'Dallas'
```

```
AND Price < 300000
```

```
AND IsActive = 1;
```

(If this returns any rows after applying a min price of 300k in UI, there is a backend or indexing defect.)

SQL Query 2

Goal: Ensure all active listings have essential values if shown in UI.

```
SELECT ListingID, City, Price, Bedrooms  
  
FROM Listings  
  
WHERE (Price IS NULL OR Bedrooms IS NULL)  
  
AND IsActive = 1;
```

SQL Query 3

Goal: Confirm backend returns listings with at least a minimum bedroom value.

```
SELECT ListingID, City, Price, Bedrooms  
  
FROM Listings  
  
WHERE City = 'Houston'  
  
AND Bedrooms < 3  
  
AND IsActive = 1;
```

IMPROVEMENT RECOMMENDATIONS

1. Add deterministic data attributes for key UI elements. Introducing attributes such as data-test-id would stabilize automation and reduce locator maintenance when UI changes.

2. Implement server-side validation for price range and filter parameters. Ensure consistent enforcement and prevent incorrect or contradictory results when invalid ranges are submitted.
3. Enhance filter state persistence using URL query parameters. Preserves consistency across pagination, browser navigation, sharing links, and improves debugging for both QA and engineering.